



CANDIDATE NAME

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CT GROUP

17S7 __

CENTRE NUMBER

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INDEX
NUMBER

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BIOLOGY

9744/04

Paper 4 Practical

28 August 2018

No additional materials are required.

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

There are **three** question booklets (I to III) to this paper. Write your **name**, **CT group**, **Centre number** and **index number** in the spaces provided at the top of this cover page and on the lines provided at the top of the cover pages of Booklets **II and III**.

Answer **all** questions in the spaces provided on the question paper.

Shift
Laboratory

INFORMATION FOR CANDIDATES

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

For Examiners' Use	
1	/ 22
2	/ 14
3	/ 19
Total Mark	/ 55

QUESTION 1

Watermelon, *Citrullus lanatus*, is a plant species in the family Cucurbitaceae, which is cultivated for its fruit. Watermelon is known to contain the enzyme, catalase.

Catalase decomposes hydrogen peroxide, H_2O_2 , into water and oxygen. H_2O_2 is a by-product of cellular respiration. As H_2O_2 is harmful, cells synthesize the enzyme catalase to remove H_2O_2 .

You are required to investigate the activity of catalase extracted from the watermelon fruit.

You are provided with:

labelled	contents	hazard		volume / cm^3
A	alginate solution	none		25
C	calcium chloride	none		40
W	distilled water	none		500
H	5% hydrogen peroxide	irritant		25
S	sulfuric acid	irritant		5
P	potassium manganate (VII) solution	harmful		25

*You are advised to wear suitable eye protection, especially when using the sulfuric acid, **S**. If **either S or P** comes into contact with your skin, wash off with cold water immediately. **P** may stain your skin.*

Proceed as follows.

- Using a fruit knife, cut the piece of watermelon given into smaller pieces on a white tile.
- Place the watermelon pieces into a mortar. Use the pestle to crush and grind the watermelon pieces to obtain a watermelon mash.
- Place a sieve on top of the opening of a large plastic vial. Filter the watermelon mash through the sieve to obtain the extract. Discard the pulp in the plastic bag provided and wash the sieve.
- To a medium plastic vial, add 6 cm^3 of watermelon extract and 6 cm^3 of **A**. Stir the mixture well with a glass rod. Ensure that you do not create air bubbles in the watermelon-alginate mixture.
- Using a retort stand, clamp a 1 cm^3 plastic dropper that has the top of its bulb cut off. Uncap the plastic vial labelled **C**. Place it beneath the plastic dropper. Pour the watermelon-alginate mixture through the plastic dropper to make watermelon-alginate beads. You should use up all the 12 cm^3 of watermelon-alginate mixture. Request for a new dropper if the watermelon-alginate mixture does not flow out of the dropper into the calcium chloride solution.

- 6 Remove the plunger from a clean and dry 10 cm³ syringe. Cover the nozzle with a piece of blu tack. Ensure that the blu tack is fitted securely onto the nozzle such that the liquid does not leak out.
- 7 Using a pair of forceps, lay the muslin cloth over the inner base of the 10 cm³ syringe.
- 8 Using a sieve, separate the watermelon-alginate beads from solution **C**.
- 9 Using a spatula, transfer watermelon-alginate beads to the syringe barrel, up to the 8 cm³ mark. Do not put the plunger back. This is the watermelon-alginate bead column.
- 10 Remove the plastic dropper from the retort stand and clamp the watermelon-alginate bead column to it. Place a small plastic vial beneath the column.
- 11 You are to fill up the watermelon-alginate bead column with sufficient 5% H₂O₂, **H**.
- (a) Describe the steps to be taken for determining the exact volume of **H** required to fill the column to the 10 cm³ mark.

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.....

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..... [2]

- 12 Using the steps you described in (a), add enough **H** to fill the column and incubate for 15 minutes. While incubating, continue with step **13** to step **15** on pages 4 and 5.
- (b) Record the volume of **H** added into the column.

volume of **H** cm³ [1]

The activity of catalase can be measured by the extent at which H_2O_2 is decomposed. The concentration of H_2O_2 present can be determined using the following redox reaction, where H_2O_2 changes the colour of potassium manganate (VII) solution from purple to colourless.

- 13** You are to prepare a series of known standards using serial dilution of 5% H_2O_2 , **H**. This will be used to determine the activity of catalase in watermelon in step **17** on page 5. In the serial dilution, you are to reduce the concentration of **H** by 10-fold between each successive dilution. You are to dilute **H** with distilled water, labelled **W**.

You will need to prepare 10 cm^3 of each concentration.

Fig. 1.1 shows the plastic vials you will use to make your serial dilution.

- (c)** Complete Fig. 1.1.

For each plastic vial:

- state, under the plastic vial, the **volume** and **concentration** of the H_2O_2 available for use in the investigation
- use one arrow, with a label above the plastic vial, to show the **volume** and **concentration** of **H** added to prepare the concentration
- use another arrow, with a label above the plastic vial, to show the **volume** of **W** added to prepare the concentration.

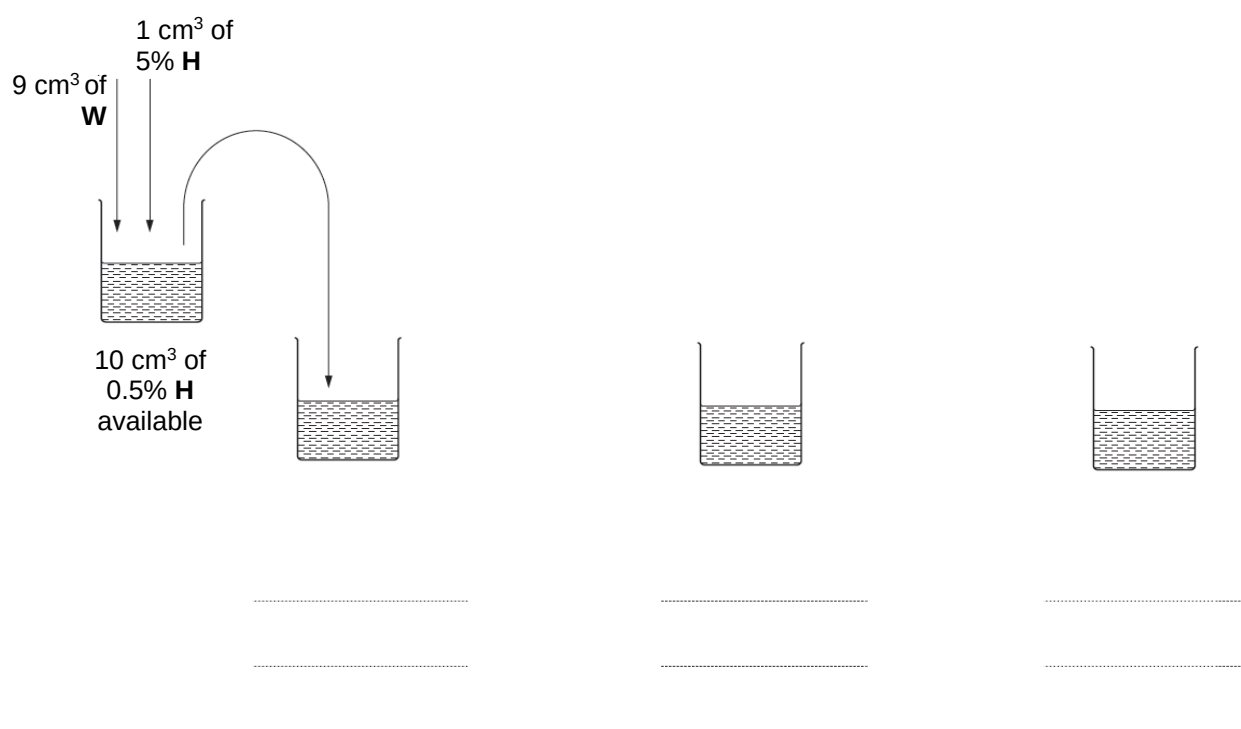


Fig. 1.1

[3]

- (d) Other than serial dilution, simple dilution can also be used.

Suggest how simple dilution can allow for more accurate results to be obtained.

[2]

- 14 On a 12-well plate, dispense 0.5 cm^3 of each dilution of **H** into an appropriately labelled well.
- 15 In a small plastic vial, add 2.5 cm^3 of **S** and 1 cm^3 of **P** to make 3.5 cm^3 of acidified potassium manganate (VII) solution. Using a 1 cm^3 dropper, add 3 drops of acidified potassium manganate (VII) solution to each well.
- (e) Record your results in a suitable table in the space below.

[3]

- 16 After 15 minutes of incubation of **H** in the watermelon-alginate column, hold the small plastic vial at the nozzle of the watermelon-alginate column. Remove the blu tack to allow the solution to be collected. This solution is known as the eluent. Discard the beads in the plastic bag provided.
- 17 Using 1 cm^3 of eluent, perform a redox reaction on the 12-well plate using the acidified potassium manganate (VII). Using the observed colour change and the results of your standards, determine the **minimum** percentage of H_2O_2 present in the eluent.

- (f) Record the minimum percentage of H_2O_2 present in the eluent.

minimum percentage of H_2O_2 [1]

- (g) Calculate the relative activity of catalase from watermelon.

Show your working clearly.

relative activity of catalase from watermelon [2]

- (h) Other than replicates and repeats, describe and explain **one** modification to the method that will increase confidence in the results obtained.

.....

 [2]

Different plant materials show varying amounts of catalase activity and the most metabolically active tissues show the greatest activity. A student carried out an investigation to:

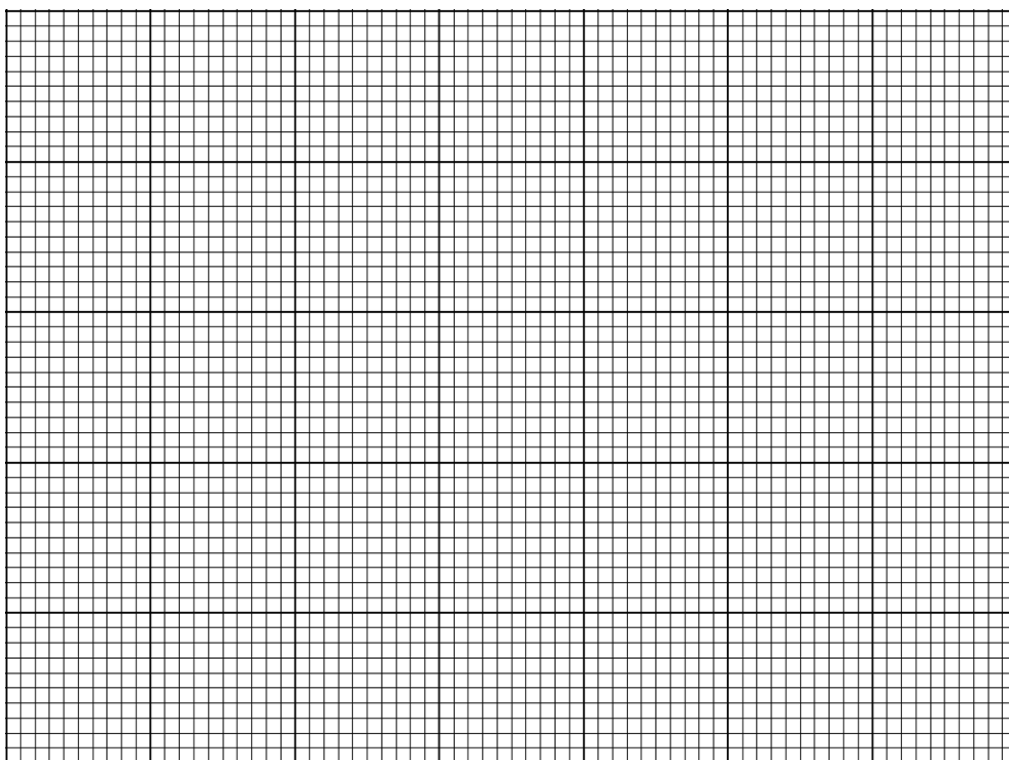
- compare the catalase activity of two different fruits of the Cucurbitaceae family
- observe the effects of immobilisation on catalase activity.

In this experiment, the student measured the time taken for the acidified potassium manganate (VII) to turn colourless and calculated the rate of reaction. The student's results are shown in Table 1.1. Standard deviation is represented by the symbol, s.

Table 1.1

rate of reaction / a.u. $\pm$ 2s			
catalase from cucumber		catalase from watermelon	
free enzyme	immobilised enzyme	free enzyme	immobilised enzyme
0.6 $\pm$ 0.08	0.3 $\pm$ 0.02	0.4 $\pm$ 0.04	0.2 $\pm$ 0.06

- (i) Use the grid to display the results shown in Table 1.1 in an appropriate form.



[3]

- (j) State what the standard deviation, s , shows about the results of this investigation.

.....
 [1]

- (k) The student carried out a statistical test to find out if the difference in the rate of reaction between the immobilised and free enzyme was significant.

The results were significant at $p < 0.05$. Explain what this means.

.....

 [2]

[Total: 22]

QUESTION 2

Both male and female mosquitoes, *Aedes aegypti* feed on plant nectars, fruit juices, and other plant sugars as their main source of energy. In addition, *Ae. aegypti* female mosquitoes feed on blood in order to produce their eggs. After feeding, the metabolic rate of these female mosquitoes increases to allow egg production to occur.

Digestion of blood in the midgut of these *Ae. aegypti* female mosquitoes results in the release of molecules like H_2O_2 that can be toxic to the mosquito. Catalase has been found to be produced during the feeding period of these *Ae. aegypti* female mosquitoes. Catalase protects these *Ae. aegypti* female mosquitoes from those toxic molecules.

A pesticide research institute, *Mosquito Squad Pte. Ltd.* isolated two forms of catalase, **P** and **Q**, from *Ae. aegypti* female mosquitoes. Catalase breaks down hydrogen peroxide into water and oxygen.

Metal ions, such as copper ions, are known to be non-competitive inhibitors of catalase. *Mosquito Squad Pte. Ltd.* is keen to investigate if copper spray is suitable to combat mosquito breeding in times of temperature elevation due to global warming. The range of temperature to be investigated is between 35 °C to 45 °C.

- (a) Using this information and your own knowledge, design an experiment to evaluate which form of catalase, **P** or **Q**, is more effectively inhibited by copper ions in times of temperature elevation.

In your plan, you must use:

- 50 cm³ of 1% of catalase **P** from *Ae. aegypti* female mosquitoes
- 50 cm³ of 1% of catalase **Q** from *Ae. aegypti* female mosquitoes
- 50 cm³ of 5% hydrogen peroxide
- 0.2% copper sulfate solution
- 10 cm³ graduated gas syringe attached to delivery tube
- thermostatically-controlled water bath and a thermometer.



You may select from the following apparatus and plan to use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes and pipette fillers, glass rods, etc.
- syringes
- timer, e.g. stopwatch.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[11]

(b) Based on your answer in **2(a)**,

- (i)** suggest a variable that was not taken into consideration in your plan when studying the catalase activity in *Ae. aegypti* female mosquitoes.

.....
..... [1]

- (ii)** identify and explain one possible biological impact of using the copper spray to combat mosquito breeding in the natural environment.

.....
.....
.....
..... [2]

[Total: 14]

QUESTION 3

During this question you will require access to a microscope, a stage micrometer, slides **S1** and **S2**.

You are not expected to be familiar with the specimens in slides **S1** and **S2**.

- (a) Slide **S1** is a stained transverse section of a stem belonging to the plant species *Cucurbita pepo*.

Examine slide **S1** under the low-power objective lens of your microscope.

- (i) You are required to use a sharp pencil for drawing.

Use the space provided to draw a plan diagram of part of the stem on slide **S1**, as shown in the shaded area of Fig. 3.1. A plan diagram only shows the arrangement of the different types of tissues. Individual cells must **not** be drawn in plan diagrams.

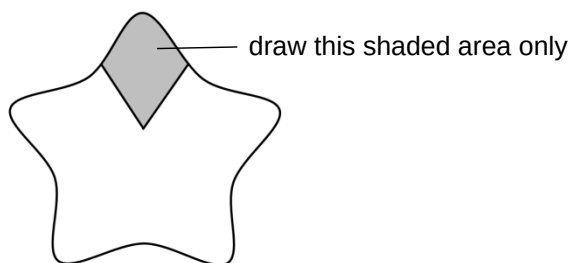


Fig. 3.1

[3]

- (ii) Observe the epidermal layer of the transverse section of the *Cucurbita pepo* stem in slide **S1**.

Using both the low-power and high-power objective lenses, choose the lens that is most suitable for determining the thickness of the epidermal layer.

State which objective lens you have decided to use.

choice of objective lens used [1]

- (iii) Using the eyepiece graticule fitted in the eyepiece lens of your microscope, and the stage micrometer, calculate the actual thickness, in μm , of the epidermal layer observed in slide **S1**.

Show your working clearly.

actual thickness of epidermal layer of **S1**, μm [3]

- (b) Slide **S2** contains two stained transverse sections of the *Cucurbita pepo* root. The slide shows actively dividing cells in the zone of cell division.

Under high-power objective lens, examine a region in either one of the two transverse sections of slide **S2**, where cells display different stages of mitosis.

Make large labelled drawings of **two** cells showing chromosomes undergoing **different** stages of mitosis.

stage of mitosis

stage of mitosis

[4]

- (c) You are required to calculate the mitotic index of cells in slide **S2**.

The number of cells undergoing mitosis can be counted to determine the mitotic index as:

$$\text{mitotic index} = \frac{\text{number of cells in mitosis}}{\text{total number of cells}}$$

- (i) Count the total number of cells in a field of view.

total number of cells [1]

- (ii) Count the number of cells that are in each stage of mitosis as observed in the same field of view seen in (c)(i).

Record your results in a suitable table in the space below.

[3]

- (iii) Calculate the mitotic index. Show your workings clearly.

mitotic index [2]

- (d) Fig. 3.2 is a photomicrograph of a stained longitudinal section through the **same** *Cucurbita pepo* root shown in slide **S2**.

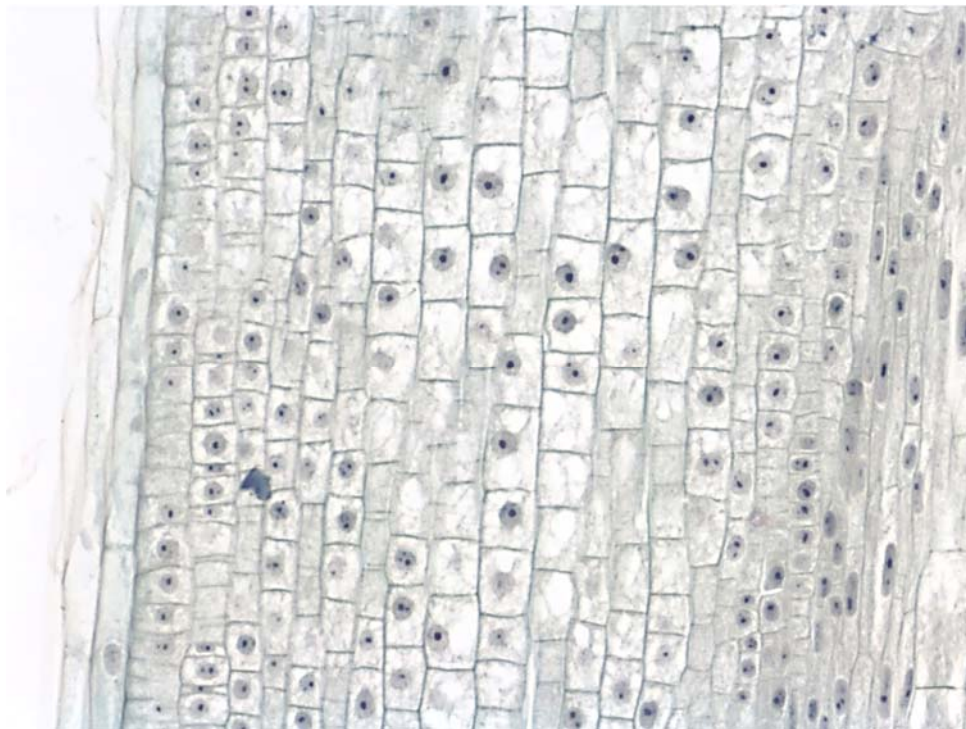


Fig. 3.2

Describe two ways in which the cells observed in Fig. 3.2 differ from those in slide **S2**.

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..... [2]

[Total: 19]

--- END OF PAPER ---